

(M, ω) compact connected symplectic manifold

goal:

create (complex) line bundles $L \rightarrow M$ that capture the symplectic geometry of M in a “linear way”. Such line bundles will be called prequantum.

what properties should we expect:

“inverse” symplectic reduction:

recall:

Let G be a Lie group acting on M preserving the symplectic form ω . A moment map for the action of G on M is a smooth map $\mu : M \rightarrow \mathfrak{g}^*$ such that:

(1) $d\langle \mu, X \rangle = X^\# \lrcorner \omega$. (the infinitesimal action of G encodes $\dim G$ many conserved quantities) for all $X \in \mathfrak{g}$.

(2) $\mu \circ \psi_g = \text{Ad}_g^* \circ \mu$, where Ad^* is the coadjoint action of G on $\mathfrak{g}^*$, ψ_g action of g (μ respects the action of G).

thm (Marsden-Weinstein, Meyer)

Let G act on M freely properly and preserving the symplectic form. Suppose $t \in \mathfrak{g}^*$ is a regular value of μ , and suppose G acts freely on $\mu^{-1}(t)$. Then $\mu^{-1}(t)/G$ is a symplectic manifold whose symplectic form ω_t satisfies $\pi^* \omega_t = \omega$.

Example

Let $\omega = \frac{i}{2} \sum_j dz^j \wedge d\bar{z}^j$ be the standard symplectic form on $\mathbb{C}^n$, $G = S^1$ (so $\mathfrak{g}^* \cong \mathbb{R}$) act in the usual way, and

$$\mu(z) = -\frac{|z|^2}{2} + \frac{1}{2}$$

Then $\mu^{-1}(0) = S^{2n-1}$, and $\mu^{-1}(0)/S^1 = S^{2n-1}/S^1 = \mathbb{CP}^{n-1}$, and you can compute that the Fubini-Study form defined chartwise by $\omega_{FS} = \frac{i}{2} \partial \bar{\partial} \log(|z|^2 + 1)$ (and patched together on the overlaps) is the pullback of ω under π .

Now, in a nbhd of a regular value $t_0 \in \mathfrak{g}^*$, we can get a 1-parameter family of symplectic quotients, and look at their cohomology:

thm (Duistermaat-Heckman)

Let S^1 act freely in a Hamiltonian way on a nbhd of $\mu^{-1}(t_0)$, then

$$[\omega_t] = [\omega_{t_0}] + \langle t - t_0, c \rangle$$

where c is the Chern class of the S^1 principal bundle $\mu^{-1}(t) \rightarrow \mu^{-1}(t)/S^1$.

(This is proved, roughly speaking, by looking at each local trivialization of the principal S^1 bundle, which induces a canonical choice of connection 1-forms based on ω (roughly, dual to ω by our moment map). Then, we patch these together, compute the curvature, then descend to cohomology).

Thus, (taking associated complex line bundles by the action of S^1 on $\mathbb{C}$), to look for “inverse” spaces of symplectic reduction (build the total space of a line bundle whose symplectic reduction is our original space), we want to find one-parameter families whose cohomology behave in this linear fashion. We will see prequantum line bundles satisfy a cohomology condition like this: the symplectic form on the reduced space of the line bundle (determined by the chern class of the bundle) will be have cohomology class that is a multiple of the original symplectic form.

geometric quantization

The next main idea is to push “prequantum line bundles linearize our symplectic manifold further”. First, we build a prequantum line bundle $L \rightarrow M$, then (picking some natural metric), we look at its space of L^2 sections $\Gamma(L)$. This gives us a Hilbert space, whose behaviour should encode symplectic geometry of M based on how we construct L . From there, quantization is a map that “distinguishes half

the coordinates of M on $\Gamma(L)$ ", motivated by physics (distinguish momentum from position) and the fact that $\Gamma(L)$ itself is "too big", and representation theoretic methods don't apply the way one would want.

This further encodes the idea of quantization commutes with reduction:

they conjectured that the following diagram commutes:

$$\begin{array}{ccc} (M, L) & \xrightarrow{\text{quantization}} & Q(L) \\ \downarrow \text{reduction} & & \downarrow \text{reduction} \\ (M_\nu, L_\nu) & \xrightarrow{\text{quantization}} & Q(L_\nu). \end{array}$$

s appear when the manifold M is no longer supposed to be

Guillemin and Sternberg proved this ('82, Inventiones) for compact Kähler manifolds, lots of people followed up and made a whole industry out of it, the main difficulty is really writing down what the map Q should be, and in less nice contexts than compact Kähler manifolds, need a whole beast of virtual representation theory. But in this sense, quantization really is inverse symplectic reduction: it gives us a "quantum" Hilbert space that we can "symplectically reduce" to get something symplectic, which we can think of as $\hbar \rightarrow 0$ in the classical limit in quantum mechanics.

When things are compact, and with the geometric quantization "reducing its size", the space of sections $\Gamma_Q(L)$ is usually finite dimensional, and one can use things like Riemann-Roch to compute useful information about it.

Three Examples

Let $M = T^*\mathbb{R}^n$ (coordinates (q, p)), and consider the trivial line bundle $L = M \times \mathbb{C} \rightarrow M$, which has canonical connection $\nabla f = df + \theta f$, $\theta = \sum_j p_j dq_j$ whose curvature is the standard ω on $T^*\mathbb{R}^n$. Then, $\Gamma(L) = L^2(\mathbb{R}^{2n})$.

The hyperplane line bundle $\mathcal{O}(1) \rightarrow \mathbb{CP}^n$, where $\mathbb{CP}^n$ is symplectic with the Fubini-Study form, where this has a Chern connection, whose curvature is ω_{FS} . This is really the main example: for a compact Kähler manifold with prequantum line bundle, we can Kodaira embed it into projective space, and then its prequantum line bundle (to a high enough tensor power) is the pullback of a hyperplane bundle AND has a nonzero section, so the quantization is meaningful.

The example I care about:

The moduli space of flat connections over a Riemann surface. Let Σ be a Riemann surface with positive genus. The space of G -connections $\mathcal{A}$ (for G a simply connected compact Lie group) (and identifying $T_{\nabla} \mathcal{A} \cong C^\infty(\Sigma, \mathfrak{g})$) has a symplectic form $\omega(\alpha, \beta) = \int_{\Sigma} \text{tr}(\alpha \wedge \beta)$ (implicitly pairing two connection 1-forms together with an Ad-invariant metric on $\mathfrak{g}$). Then the gauge group $\mathcal{G} = C^\infty(\Sigma, G)$ acts in a Hamiltonian way on $\mathcal{A}$ whose moment map takes the curvature of a connection, so there is a reduced space $\mu^{-1}(0)/\mathcal{G}$ which is the moduli space of flat connections. Then, this has a prequantum line bundle as follows: there is a functional $\Theta : \mathcal{A} \times \mathcal{G} \rightarrow \mathbb{C}$ by

$$\Theta(A, g) = \exp i(CS(gAg^{-1}) - CS(A))$$

where

$$CS(A) = \frac{1}{4\pi} \int_N \text{tr}(A \wedge dA + \frac{2}{3} A^3)$$

where N is an arbitrary 3-manifold with boundary Σ . We can define a line bundle (the Chern-Simons line bundle) as follows: $\mathcal{G}$ acts on $\mathcal{A} \times \mathbb{C}$ by $g : (A, z) \mapsto (gAg^{-1}, \Theta(A, g)z)$, and so we can take the associated line bundle

$$L = \mu^{-1}(0)/\mathcal{G} \times_{\mathcal{G}} \mathbb{C}$$

(where associated line bundle is take the quotient of $\mathcal{A} \times \mathbb{C}$ under this action by $\mathcal{G}$). This turns out to be prequantum (hard computation).

Definition and Existence

Definition NON-STANDARD

A prequantum line bundle is:

- (1) A Hermitian line bundle $L \rightarrow M$
- (2) Equipped with a faithful unitary representation of Lie algebras ($\mathcal{C}^\infty(M)$ with Poisson bracket), $Q : \mathcal{C}^\infty(M) \rightarrow \mathfrak{u}(\mathcal{H})$, where $\mathcal{H} = \Gamma(L)$, which:
 - (2a) Q_f depends only on the first order behaviour of $f \in \mathcal{C}^\infty(M)$.
 - (2b) Q_f lifts the Hamiltonian vector field assignment, in the sense that

$$Q_f(s \cdot \psi) = (X_f s)\psi + sQ_f \psi$$

for all $f, s \in \mathcal{C}^\infty(M)$ and $\psi \in \mathcal{H}$. In essence, we extend the domain of X_f to include $\mathcal{H}$.

Lemma

There is a unitary connection ∇ on $L \rightarrow M$ and a nonzero constant $c \in \mathbb{R}$ so that $Q_f = \nabla_{X_f} + icf$.

Proof:

Step 1: if $s \in \mathcal{C}^\infty(M)$ is constant, then $Q_s(\psi) = ics\psi$ for some $c \in \mathcal{C}^\infty(M)$.

By the Hamiltonian lifting property, the operator $Q_1 : \mathcal{H} \rightarrow \mathcal{H}$ is $\mathcal{C}^\infty(M)$ -linear. Since Q_1 is unitary, this forces $Q_1(\psi) = ic\psi$ for some $c \in \mathcal{C}^\infty(M)$. The result then follows by linearity.

Step 2: If $f, g \in \mathcal{C}^\infty(M)$ are such that $X_f = X_g$, $f - g$ is constant, so part (1) shows $Q_f - icf = Q_g - icg$, so $\nabla : \mathfrak{X}_H(M) \times \mathcal{H} \rightarrow \mathcal{H}$, $\nabla_{X_f}(\psi) = Q_f - icf$ is well-defined. The Hamiltonian lifting property shows its a derivation.

Step 3: To show its a unitary connection, if X_f vanishes at $x \in M$, then df vanishes at x . Let $c \in \mathcal{C}^\infty(M)$ be the constant with value $f(x)$. By the first-order condition, $Q_{f-c}\psi$ vanishes at x for all ψ . Therefore, $\nabla_{X_f}\psi = Q_{f-c} - ic(f-c)$ vanishes at x for all ψ , so ∇ is tensorial in X_f (by the usual bump function arguments), so the map $\psi \mapsto (\nabla_{X_f}\psi)(y)$ depends only on ψ and the value of X_f at y . Then, unitarity follows from the unitarity of Q .

Step 4: c is a nonzero constant.

Since Q is a homomorphism of Lie algebras, we have

$$0 = Q_{\{f,1\}}\psi = [\nabla_{X_f} + icf, ic]\psi$$

and so

$$0 = [\nabla_{X_f}, c]\psi = (\nabla_{X_f}c)\psi$$

for all $f \in \mathcal{C}^\infty(M)$ and $\psi \in \mathcal{H}$. By connectedness, c is constant, and it must be nonzero because Q is faithful. ■

Thus, the connection splits into the horizontal part ∇_{X_f} and the vertical part icf .

Lemma

The curvature of the connection ∇ is $F^\nabla = ic\omega$.

Proof:

$$\nabla_{X_{\{f,h\}}} + ic\{f,h\} = Q_{\{f,h\}} = [Q_f, Q_h] = [\nabla_{X_f} + icf, \nabla_{X_h} + ich] = [\nabla_{X_f}, \nabla_{X_h}] + 2ic\{f,h\}$$

where the last equality is because $[\nabla_{X_f}, ich]\psi = ic\{f,h\}\psi$ for all ψ . Rearranging,

$$ic\omega(X_f, X_h) = [\nabla_{X_f}, \nabla_{X_h}] - \nabla_{[X_f, X_h]} = F^\nabla(X_f, X_h)$$

Standard Definition

A prequantum line bundle is:

- (1) A Hermitian line bundle $L \rightarrow M$
- (2) A unitary connection with curvature $F^\nabla = ic\omega$ for some nonzero constant.

Theorem (Kostant-Souriau, '70)

A compact connected symplectic manifold admits a prequantum line bundle iff $c[\omega] \in H^2(M)$ is integral.

Proof:

Because $F^\nabla = ic\omega$, and the Chern class $[F^\nabla]$ is integral, we obtain one direction.

Conversely, write down a bunch of Čech cocycles and definition chase, then use the de Rham-Čech isomorphism. ■

Moreover, writing the Čech cocycles down more intensely, such a prequantum line bundle is more or less unique.